

Clonezilla tool



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1. General description of Clonezilla

Clonezilla is a versatile program specifically designed for disk cloning, making it an essential tool in system development, backup, and recovery processes. Notably, it offers the advantage of saving and restoring only the used blocks on a hard disk, optimizing the cloning process and reducing storage requirements.

One of the strengths of Clonezilla is its extensive support for various file systems. It can work seamlessly with file systems such as ext2, ext3, ext4, reiserfs, reiser4, xfs, jfs, btrfs, f2fs, and nilfs2 in GNU/Linux. Moreover, it is compatible with FAT12, FAT16, FAT32, exFAT, and NTFS in MS Windows, HFS+ and APFS in Mac OS, UFS in FreeBSD, NetBSD, and OpenBSD, minix in Minix, as well as VMFS3 and VMFS5 in VMWare ESX. This broad range of file system support makes Clonezilla a versatile tool that can be used across different operating systems, including Windows, macOS, and Linux.

In addition to its extensive compatibility, Clonezilla boasts another noteworthy feature: it is an open-source software distributed under the GNU General Public License (GPL) Version 2. Being open source means that Clonezilla is freely available, allowing users to utilize its powerful disk cloning capabilities without incurring any monetary costs. This aspect contributes to its popularity and widespread adoption worldwide.

Clonezilla finds application in various use cases across different sectors. For instance, in enterprise environments, it is commonly employed for disk cloning in product testing scenarios. In educational settings, it is used to clone USB flash drives, each containing an operating system and associated data. These are just a few examples of the numerous ways in which Clonezilla is leveraged to streamline disk cloning tasks in diverse fields. [1]

2. The main features of Clonezilla:

1. **Disk Cloning and Imaging:** Enables the creation of exact copies of entire disks or specific partitions, useful for full system restores, upgrades, or deployment across multiple machines.
2. **Multicast Support:** Utilizing multicast technology, the software can clone to multiple machines simultaneously, streamlining the setup process in organizations.
3. **Different Modes of Cloning:** Offers both device-to-device cloning, for direct disk copying, and device-to-image cloning, for creating a restorable image file.
4. **Bootable Media:** Accessible via a bootable CD/DVD or USB flash drive, it functions independently of an operating system, crucial for restoring systems without an OS.
5. **Broad Filesystem Support:** Compatible with a variety of filesystems including ext2, ext3, ext4, fat16, fat32, ntfs, accommodating diverse operating systems like Windows, macOS, and Linux.
6. **Differential / Incremental Backup:** Complements complete disk images with support for differential and incremental backups, reducing space requirements for ongoing backups.
7. **Compression and Encryption:** Features image compression for space efficiency and supports AES-256 encryption for data security.

8. **Broad Hardware Support:** It can work with various hardware, including BIOS and UEFI machines, and is compatible with a wide array of storage devices like HDDs, SSDs, and various RAID configurations.
9. **Flexible Restoration:** You can restore an image to a new disk or partition that is either larger or smaller than the original, as long as the data fits on the new one.
10. **Customizable Scripts:** Advanced users can create custom scripts to automate tasks in Clonezilla, making it easier to deploy in large-scale environments.
11. **Network Support:** Facilitates saving and restoring images over networks using protocols like SSH, Samba, NFS, and WebDAV.
12. **Rescue Mode:** Includes a mode for rescuing broken systems.
13. **Accessibility:** The interface, accessible via Live CD, ensures usability on machines with limited resources.
14. **Partition Table Manipulation:** Capable of handling partition tables, allowing for exact duplication of partition layouts.

These features make Clonezilla a very flexible tool for both individual and enterprise users who need a reliable solution for system backup, deployment, and recovery. Its versatility and powerful cloning capabilities, combined with the cost benefit of being free, make it a popular choice among IT professionals. [2]

3. Comparing Clonezilla features with similar tools:

1. Clonezilla vs. Acronis True Image:

- Clonezilla is an open-source tool, while Acronis True Image is a commercial product with a free trial.
- Acronis True Image provides more user-friendly features and a graphical user interface (GUI), making it easier for non-technical users.
- Acronis True Image offers a wide range of backup options, including full image backups, file-level backups, and continuous data protection. [3]

2. Clonezilla vs. Duplicati:

- Clonezilla is primarily used for creating disk images, while Duplicati is focused on file and folder backups.
- Duplicati supports various backup destinations, including local drives, network drives, and cloud storage providers, making it more versatile for offsite backups.
- Clonezilla offers complete system backup capabilities, while Duplicati is better suited for regular file backups with features like encryption and deduplication.

3. Clonezilla vs. Macrium Reflect:

- Macrium Reflect is a commercial backup software with a free edition, while Clonezilla is open-source and free.
- Macrium Reflect provides faster backup and recovery options and has a user-friendly interface.
- Clonezilla is more suitable for advanced users who want to create complete disk images, while Macrium Reflect caters to a broader range of users with different backup needs.

4. Clonezilla vs. Amanda Network Backup:

- Clonezilla is primarily a local backup tool, whereas Amanda Network Backup is designed for network-based backup solutions.
- Amanda Network Backup is suitable for enterprises that need to back up multiple computers to a central server, whereas Clonezilla is better for individual system backups. [4]

4. An example of how to use Clonezilla

Clonezilla for Hard Disk Cloning

It should be noted that the external drive for data backup must be at least the same size as the drive for disk cloning. You won't have to waste time stopping to hunt for a bigger disk if you do that Use a bootable USB device or the Clonezilla Live CD to start your system. [5]

Step 1: Download Clonezilla from the official website by clicking the provided [link](#).

Step 2: Next, make sure the version you plan to download is compatible with your system and stable. Later on, you will need to either make a CD/DVD of Clonezilla or a bootable USB stick.

Step 3: To backup your data, you must connect both the external disk and the bootable device containing Clonezilla. When you're sure they've been fitted correctly, please restart the system.

Step 4: Choose to boot from Clonezilla disk, which will subsequently prompt you with Clonezilla welcome screen. Select the option Clonezilla live option and press enter to continue.

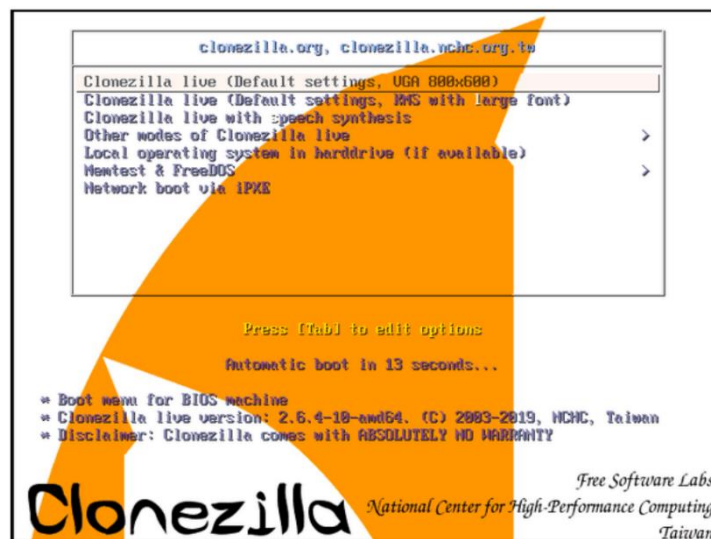


Figure 1: Clonezilla Welcome Screen

Step5: The language screen will then show up. Here, you must choose the language that is appropriate for the cloning procedure.

The Up and Down arrow keys come in helpful for sifting through the many possibilities. Following the Linux system's RAM filling, this phase takes place.

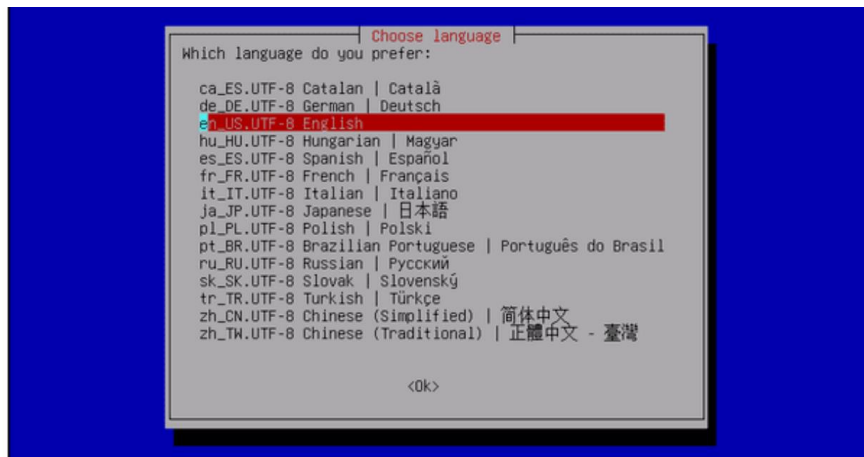


Figure 2: Language Screen

Step 6: You have the ability to customize or alter your keyboard layout in the following step. The versions you have here may vary according on the type of Clonezilla you want to utilize. The more recent iterations of Clonezilla, however, offer just two options.

The first is the US keyboard layout that is used by default, while the second is the keyboard layout changing option. We shall choose the first choice in this instance. Select "don't touch keymap" if you are using an earlier version of Clonezilla.

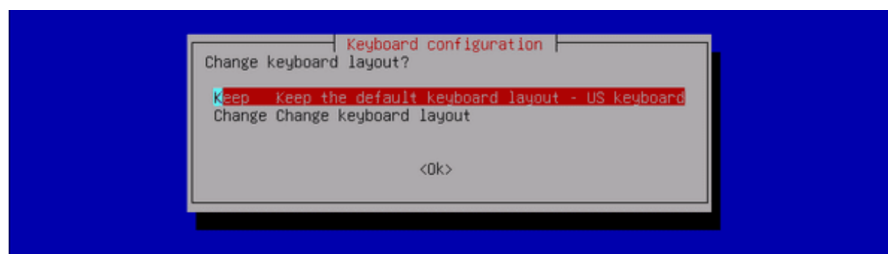


Figure 3: Configure Keyboard Layout

Step 7: The start screen then shows up, giving you the choice to launch Clonezilla or dive right in and start interacting with the shell. In order to continue, we choose the option Start Clonezilla and hit Enter.

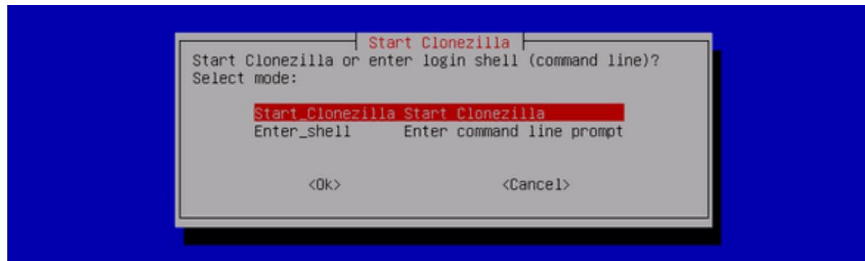


Figure 4: Clonezilla Start Screen

Step 8: Selecting the device-image alternative and pressing Enter will allow us to clone a local encrypted disk image. The resulting image of the cloned hard disk will be saved on a network space for this example.

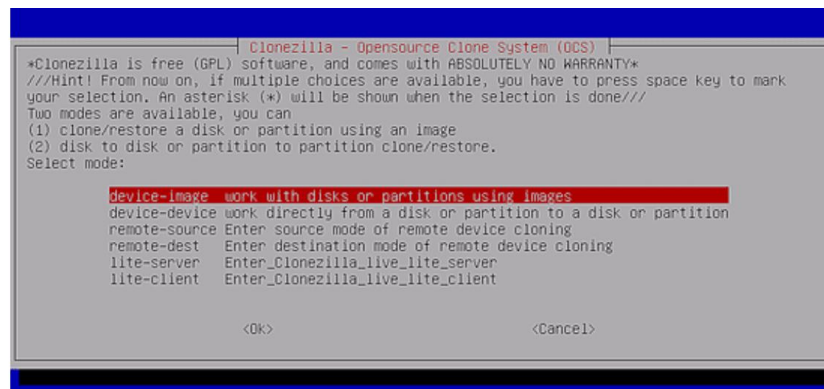


Figure 5: Use the Device-Image

Step 9: The cloned image will be saved on the network shared directory using the CIFS/SMB protocols. To store the cloned picture, you are also free to utilize other widely used network protocols.

A local connected driver, an SSH server, a WebDAV server, and an NFS share are a few examples of these protocols. We will choose samba_server from this interface and press the Enter key to proceed.

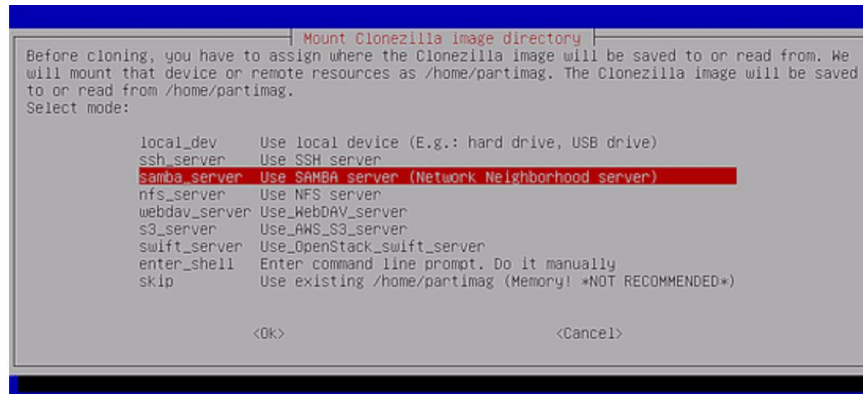


Figure 6: Samba_Server and Press Enter to Continue

Step 10: When configuring the network interface, select the DHCP option. In a given scenario, there are several network interfaces. Select the DHCP network's physically linked interface.

Select the static option to manually add the network interface settings if a DHCP server is not setup.

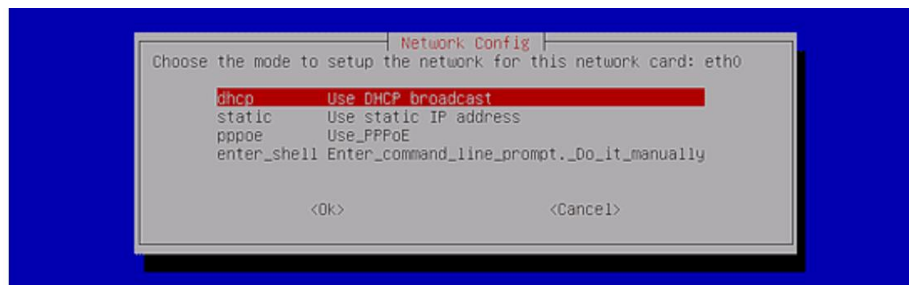


Figure 7: Select DHCP Option to Configure the Network Interface

Step 11: At this point, the Samba server's IP address is added. Alternatively, the Fully Qualified Domain Name may be this. Next, press Enter to move on to the next screen.

Include the fully qualified domain name or an IP address.

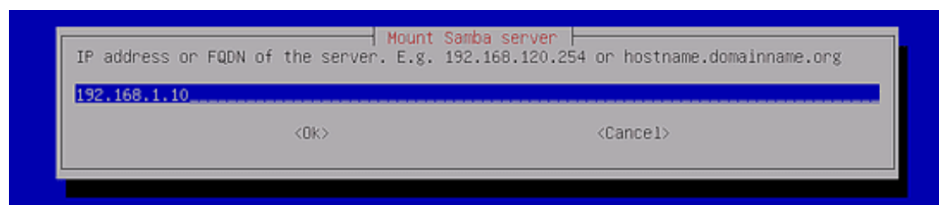


Figure 8: Add IP Address or the Fully Qualified Domain Name

Step 12: Leave the domain box empty if there isn't a recognized domain established on your Samba server. However, if it has, please note so in the designated space as shown in the image below. Next, click "enter" to continue.

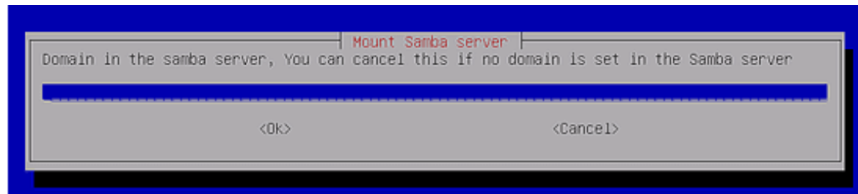


Figure 9: Leave the Domain Field Blank

Step 13: You may add the name of a Samba server account on the screen that appears next. Actually, you may grant the server write permission and hit the Enter key to proceed.

add a Samba server account name.

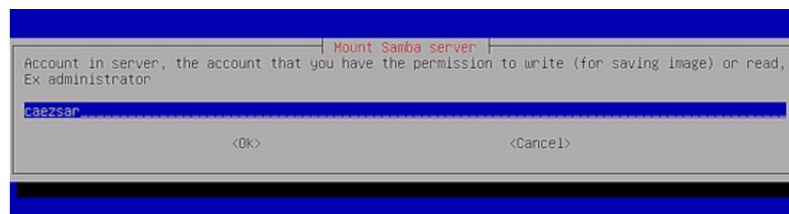


Figure 10: Add the Name of a Samba Server Account

Step 14: At this point, you specify the directory's absolute path from the Samba server. The copied picture is kept at this location.

Give the directory's absolute path from the Samba server.

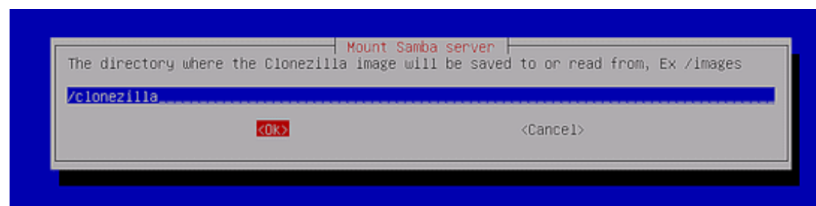


Figure 11: Specify the Absolute Path of the Directory from the Samba Server

Step 15: Choose the updated Samba protocol version. And hit "enter" to go on to the next prompt. Use protocol version 1.0 if the shared network location is hosted by an outdated SMB server.

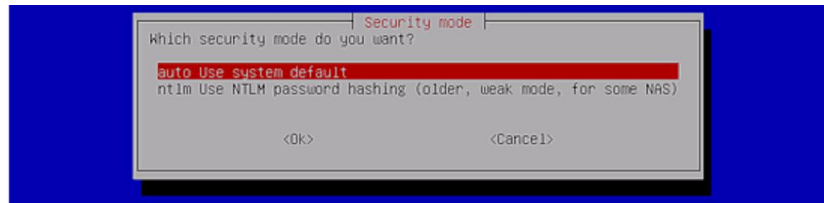


Figure 12: Auto Use System Default Security Mode

Step 17: At this point, your Samba account has to have a password added. This comes before mounting the share and moving forward by hitting the "enter" key.

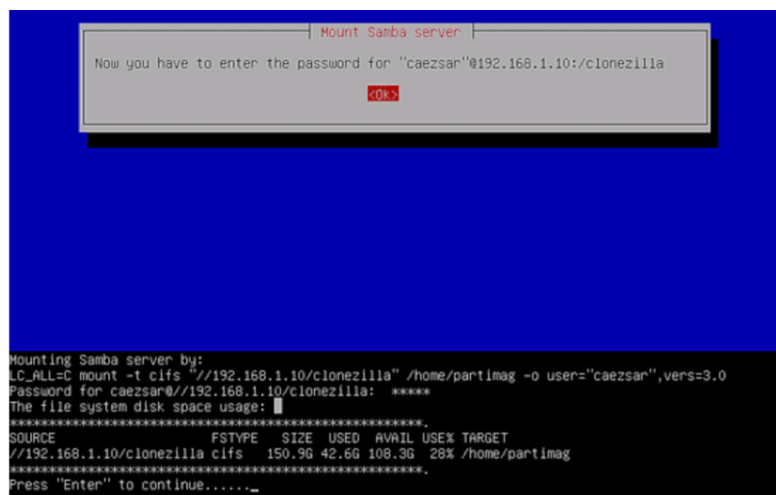


Figure 13: Add a Password for Samba Account

Step 18: The user is given the option to choose the wizard's operating mode in the subsequent interface. You can choose between basic and expert options.

Unless you modify it, the beginning mode is always chosen by default. You can choose the divisions that interest you from this point on.

Be aware that unless you are really confident in your abilities, using the expert mode calls for extra caution. You could lose your data if this isn't the case.

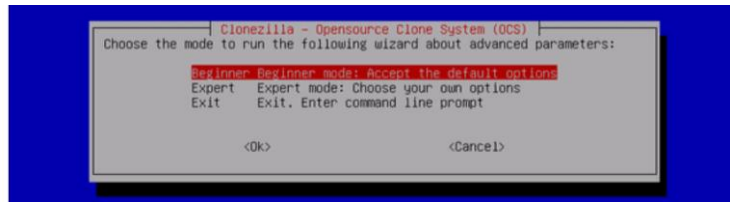


Figure 14: Choose Beginner or Expert Mode

Step 19: To image the locally attached IDE/SCI/SATA drive, select the savedisk choice.

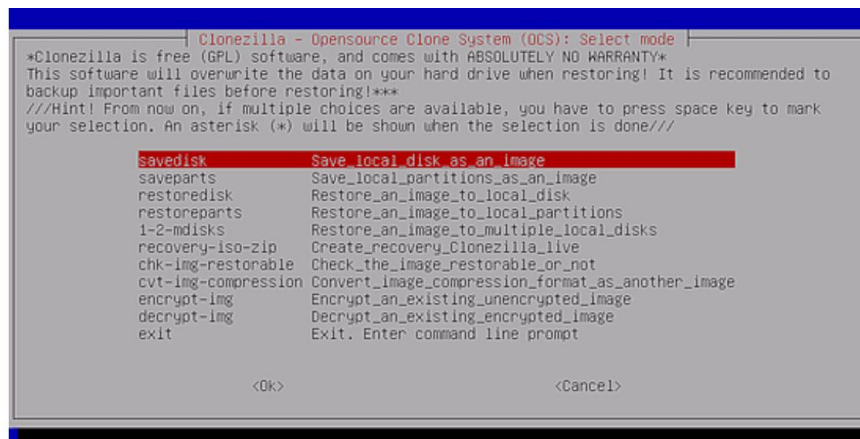


Figure 15: Select Savedisk Option

Step 20: You may either manually add a descriptive name to your stored image or keep the default naming scheme that Clonezilla adds for you.

Choosing to alter the image's name often gives you the flexibility of being able to quickly identify it in the future. It is also strongly advised that the date the photo was shot be provided.

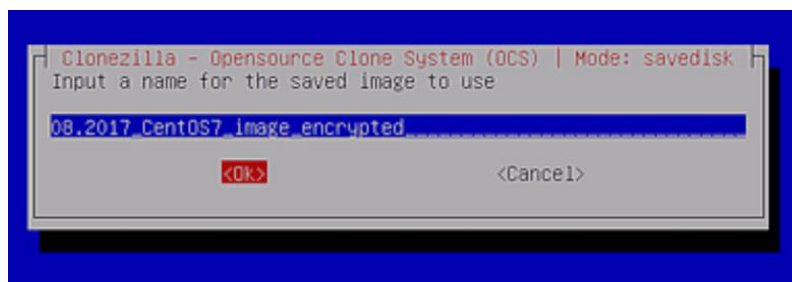


Figure 16: Choose the Default Name Scheme Clonezilla Inputs

Step 21: Select which local source disk you want to copy over. It is important to pay attention to the names used while choosing the source disk, since they may be unfamiliar to you.

The most often used conventions are sda, sdb, and so forth. If you are having trouble doing this, you may try restarting the computer and verifying your source disk via the BIOS. If you are certain in the disk's size, utilizing it is another crucial alternative.

If you run into trouble, you can obtain information by utilizing the disk's serial number to identify the correct device.

Here, we're going to clone an image using a VMware drive. Actually, use the space bar to choose the device after selecting the correct source disk. Hit the Enter key to proceed.

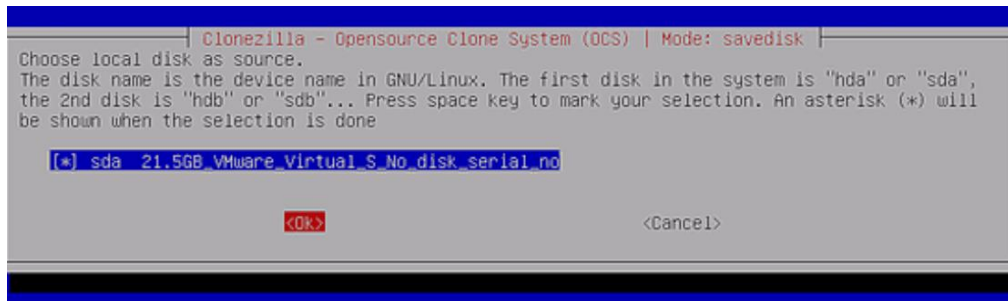


Figure 17: Select the Local Source Disk

Step 22: If you are positive that the source file system is not corrupted, choose "Skip checking/repairing source file system" and hit the enter key to continue.

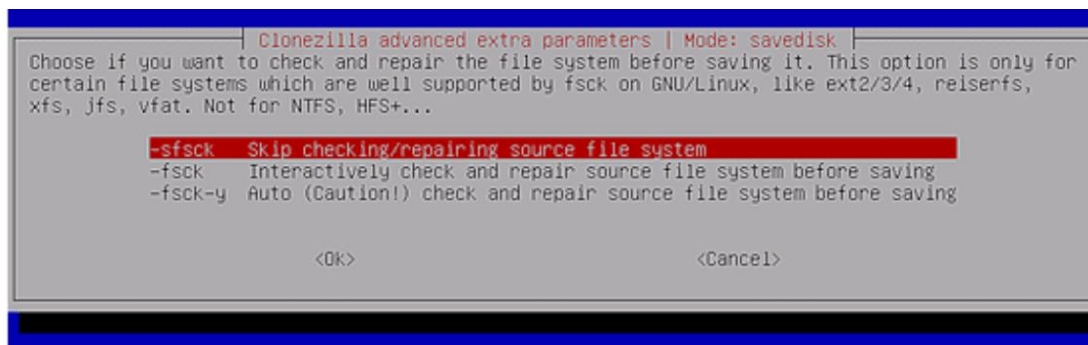


Figure 18: Skip Checking or Repairing the Source File System

Step 23: Select "no" on this screen. This will guarantee that there will be no need to inspect the stored image. Next, choose OK to proceed.

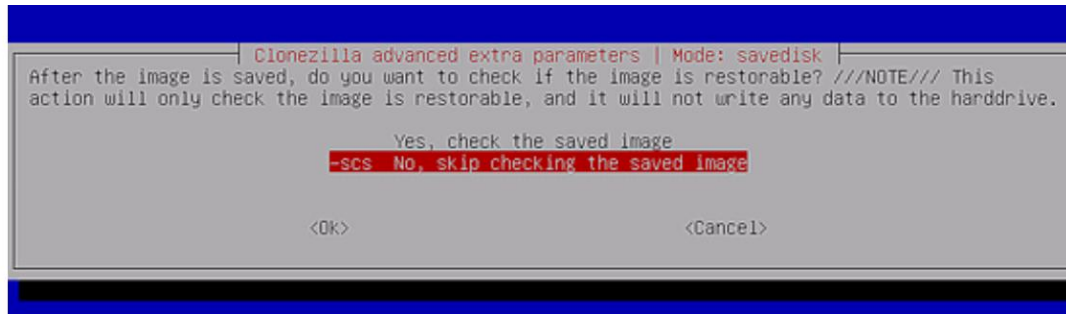


Figure 19: Select No to Check if the Saved Image is Restorable

Step 24: Select the encryption of the picture as the second option here. To proceed, choose OK and then hit "enter." Consequently, the cloned image that is produced is encrypted using the eCryptfs application.

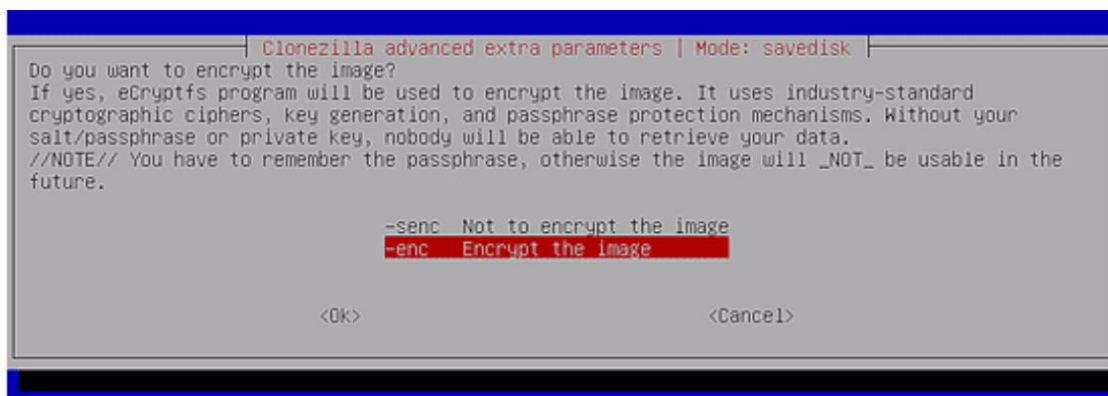


Figure 20: Encrypt the Image

Step 25: The cloning process is taken over by Clonezilla. If you choose to proceed with the procedure, it first displays a report of the activity that is taking place along with two cautions.

To indicate no, press n, and to indicate yes, press y.

Following this, a choice to clone the boot loader or not is presented in the third warning.

```
PS. Next time you can run this command directly:
/usr/sbin/ocs-onthefly -g auto -e1 auto -e2 -r -j2 -f sda -t sdb
This command is also saved as this file name for later use if necessary: /tmp/ocs-onthefly-2016-08-17-14-26
*****
Press "Enter" to continue...
*****
Searching for data partition(s)...
Excluding busy partition or disk...
Unmounted partitions (including extended or swap):
Collecting info. done!
WARNING!!! WARNING!!! WARNING!!!
WARNING! THE EXISTING DATA IN THIS HARDDISK/PARTITION(S) WILL BE OVERWRITTEN! ALL EXISTING DATA WILL
BE LOST: sdb
*****
Machine: VMware Virtual Platform
sdb (21.5GB_VMware_Virtual_S_No_disk_serial_no)
*****
Are you sure you want to continue? (y/n) y
OK, let's do it!!
*****
Let me ask you again.
WARNING!!! WARNING!!! WARNING!!!
WARNING! THE EXISTING DATA IN THIS HARDDISK/PARTITION(S) WILL BE OVERWRITTEN! ALL EXISTING DATA WILL
BE LOST: sdb
Are you sure you want to continue? (y/n) y
OK, let's do it!!
*****
Do you want to clone the boot loader (executable code area, the first 446 bytes) to: sdb ?
[y/n] y
```

Figure 21: Confirm if You Wish to Continue Cloning

Step 26: When you choose to move on with the cloning, Clonezilla first creates a partition table on the target disk that you want to use.

A warning then appears informing you whether or not to clone the boot loader. In this instance, we hit Y in order to finish the cloning procedure.

```
Trying to clean the MBR and GPT partition table on the destination disk first: /dev/sdb
Informing the OS of partition table changes..... done!
Creating MBR type's partition table...
No partition table exists in target disk /dev/sdb, try to initialize one so that we can get the disk
size by parted... Running: parted -s /dev/sdb mklabel msdos
done!
sfdisk --force /dev/sdb < /tmp/ocs_onthefly_local.M3a5hJ/tgt-pt.sf
Checking that no-one is using this disk right now ... OK

Disk /dev/sdb: 20 GiB, 21474836480 bytes, 41943040 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x97095c9e

Old situation:

>>> Script header accepted.
>>> Script header accepted.
>>> Script header accepted.
>>> Script header accepted.
>>> Created a new DOS disklabel with disk identifier 0xee0d869a.
Created a new partition 1 of type 'Linux' and of size 500 MiB.
/dev/sdb2: Created a new partition 2 of type 'Linux LVM' and of size 19.5 GiB.
/dev/sdb3:
New situation:

Device Boot Start End Sectors Size Id Type
/dev/sdb1 * 2048 1026047 1024000 500M 83 Linux
/dev/sdb2 1026048 41943039 40916992 19.5G 8e Linux LVM

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
This was done by "sfdisk --force /dev/sdb < /tmp/ocs_onthefly_local.M3a5hJ/tgt-pt.sf"
Informing the OS of partition table changes....
```

Figure 22: Create Partition Layout on the Destination Disk Drive

Step 27: The cloning process begins here without the user's interaction, therefore you must be careful not to turn off your computer at this crucial point.

This is a graphical report that shows every activity that is occurring.

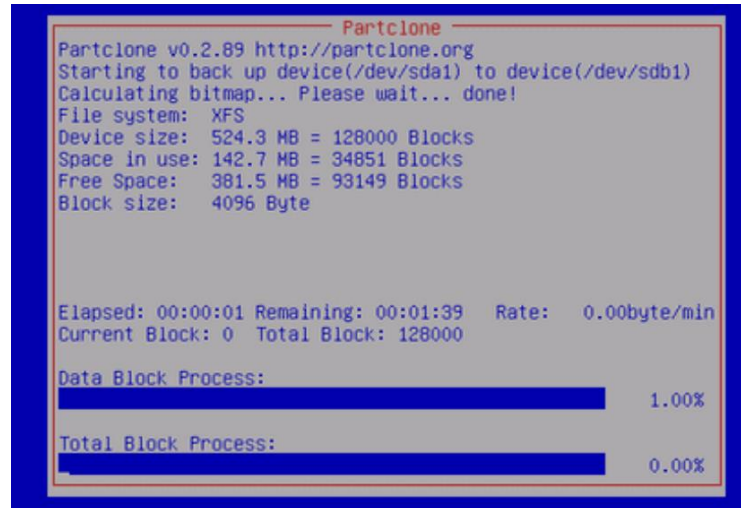


Figure 23: Disk Cloning Process By Clonezilla

Step 28: After the cloning procedure is finished, a report is produced. Additionally, you are given the choice to use Clonezilla once again. selecting option 2 will launch the exit wizard; selecting option 1 will keep you in console mode. As an alternative, use "enter" to bring up the exit screen.

```
Skip updating syslinux on that.
Device /dev/sdb2 is not a FAT partition.
Skip updating syslinux on that.
*****
Running: ocs-install-grub -s sdb auto
Found grub partition: /dev/sdb1
Found boot loader grub in the MBR of disk /dev/sdb.
There is a separate boot partition in target device. Trying to mount root partition for grub-install
to work with that...
Setting up the Logical Volume Manager
  0 logical volume(s) in volume group "centos" now active
No root partition is mounted! You'd better to make sure the running kernel does support the file sys
tem of root partition!
Found grub 2 installed in the restored OS.
Variable root_prt not assigned in function test_run_grub2_from_restored_os.
Program terminated.
*****
Try to run partclone.ntfsfixboot for NTFS boot partition if it exists. Scanning partition(s): sdb1
sdb2...
The NTFS boot partition was not found or not among the restored partition(s). Skip running partclone
.ntfsfixboot.
*****
Checking if udevd rules have to be restored...
Now syncing - flush filesystem buffers...

Ending /usr/sbin/ocs-onthefly at 2016-08-17 14:28:30 UTC...
*****
If you want to use Clonezilla again:
(1) Stay in this console (console 1), enter command line prompt
(2) Run command "exit" or "logout"
*****
When everything is done, remember to use 'poweroff', 'reboot' or follow the menu to do a normal powe
roff/reboot procedure. Otherwise if the boot media you are using is a writable device (such as USB f
lash drive), and it's mounted, poweroff/reboot in abnormal procedure might make it FAIL to boot next
time!
*****
Press "Enter" to continue..._
```

Figure 24: Cloning Disk Complete By CloneZilla

Step 29: There are four options on the screen that appears next. These consist of:

- "Poweroff" to halt.
- "reboot" to resume.
- "cmd" to launch the interactive command-line interface.
- "rerun1" to start the cloning operation over.

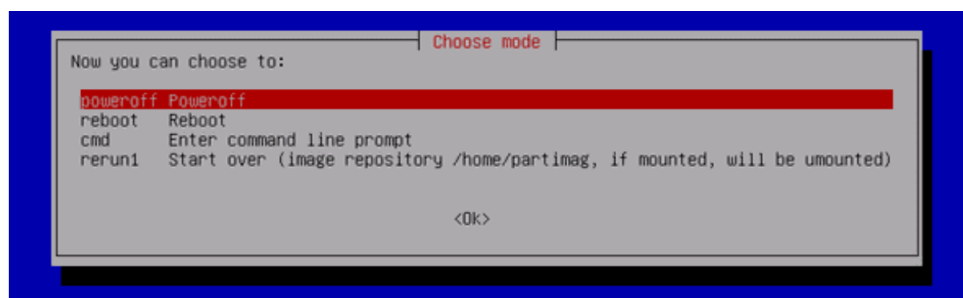


Figure 25: Final Options Presented By Clonezilla

The disk drive copying process is now complete. At this point, your only options are to disconnect your external drive, store it somewhere secure, swap out your existing disk, and boot your computer using the new disk.

5. Clonezilla advantages and disadvantages:

Advantages of Clonezilla:

- **Free and Open-Source:** Clonezilla is open-source software, which means it's free to use and can be customized or modified to suit your needs.
- **Cross-Platform Support:** It is compatible with Windows, macOS, and Linux, making it versatile and accessible to a wide range of users.
- **Disk Imaging:** Clonezilla can create complete disk images, including the operating system, applications, and data, providing a comprehensive backup solution.
- **Disk Cloning:** It allows for easy disk cloning, enabling users to replicate the entire contents of one drive to another, which can be useful for upgrading to a larger hard drive or replacing a failing one.
- **Support for Various File Systems:** Clonezilla supports a wide range of file systems, making it compatible with different storage devices and operating systems.

- **Multicast Support:** It can deploy disk images to multiple computers simultaneously over a network using multicast, which can be very efficient for deploying a standardized system to many computers at once.
- **Text-Based Interface:** While it may not be the most user-friendly interface, the text-based interface allows for scripting and automation, which can be beneficial for advanced users and system administrators.

Disadvantages of Clonezilla:

- **Complex for Beginners:** The text-based interface can be intimidating for novice users. Creating and restoring disk images with Clonezilla may require a learning curve.
- **No Real-Time Backups:** Clonezilla is primarily a tool for creating full disk images or cloning drives. It does not provide real-time or continuous backup solutions.
- **Limited Incremental Backup Support:** While Clonezilla does offer some support for incremental backups, it may not be as feature-rich or user-friendly as dedicated incremental backup tools.
- **No File-Level Backups:** Clonezilla operates at the disk or partition level and does not provide the ability to back up individual files or folders.
- **Lack of Graphical User Interface (GUI):** Clonezilla's interface is text-based, which can make it less appealing to users who prefer graphical user interfaces. However, there are alternative tools that provide a GUI for Clonezilla.
- **No Real-Time Synchronization:** Clonezilla is not designed for real-time data synchronization between devices, so it may not be suitable for scenarios where continuous data updates are critical [6].

References

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